fn then_index_or_default

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of then_index_or_default in mod.rs at commit 286ce37ce (outdated[1]).

This postprocessor indexes into a vector or returns the default value of the type if the index does not exist.

1 Hoare Triple

Precondition

Compiler-verified

- Generic T implements trait Default

Caller-verified

None

Pseudocode

```
def then_index_or_default(
    index: usize,
) -> Function[Vec[T], T]:
    return Function.new(lambda x: x[index] if index < len(x) else T.default())
```

Postcondition

Theorem 1.1. For every setting of the input parameters (index, T) to then_index_or_default such that the given preconditions hold, then_index_or_default raises an error (at compile time or run time) or returns a valid postprocessor. A valid postprocessor has the following property:

1. (Data-independent errors). For every pair of members x and x' in input_domain, function(x), function(x') either both raise the same error, or neither raise an error.

Proof. The function is infallible, so the function satisfies the data-independent errors property. Therefore the postcondition is satisfied. □

[1] See new changes with git diff 286ce37ce..58d6f416 rust/src/transformations/scalar_to_vector/mod.rs